

RaceDay — API Endpoint Plan

This plan covers every endpoint required to support the functional requirements: Authentication, User Profile, Events, Categories, Event Enrolments, and Results. It also includes supporting endpoints for Routes and Weather, which the participant-facing "prepare for race day" requirement depends on. **Role Required** key: None = public, Any = any logged-in user, Participant / Organiser = that role only, Owner (Organiser) = the organiser who created the event/category/registration being acted on.

Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as either a Participant or an Organiser.	None	{ firstName, lastName, email, password, role, phoneNumber }	201 Created — user object + auth token. 400 Bad Request — email already registered or missing fields.
POST	/api/auth/login	Authenticates a user and issues an auth token.	None	{ email, password }	200 OK — auth token + user object. 401 Unauthorized — invalid credentials.

User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the logged-in user's own profile.	Any	None	200 OK — user object. 401 Unauthorized — no/invalid token.
PUT	/api/users/me	Updates the logged-in user's own profile details.	Any	{ firstName, lastName, phoneNumber, email }	200 OK — updated user object. 400 Bad Request — invalid data.
GET	/api/users/me/registrations	Lists the logged-in participant's past and upcoming event entries.	Participant	None	200 OK — array of registrations (with event and category summary).
GET	/api/users/me/results	Returns the logged-in participant's personal performance history across all events entered.	Participant	None	200 OK — array of result records (with event and category summary).

Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists all upcoming/published events. Supports optional ?date= and ?location= filters.	None	None	200 OK — array of event objects.
GET	/api/events/{id}	Returns full details for a single event, including its categories and route summary.	None	None	200 OK — event object. 404 Not Found — event does not exist.
POST	/api/events	Creates a new event owned by the logged-in organiser.	Organiser	{ eventName, eventDate, location, description }	201 Created — event object. 400 Bad Request — missing/invalid fields.
PUT	/api/events/{id}	Updates an event's details.	Owner (Organiser)	{ eventName, eventDate, location, description, status }	200 OK — updated event object. 403 Forbidden — not the event owner. 404 Not Found.
DELETE	/api/events/{id}	Cancels/removes an event.	Owner (Organiser)	None	204 No Content. 403 Forbidden — not the event owner. 404 Not Found.
GET	/api/events/{id}/route	Returns route/course details for an event (distance, elevation, start location, GPX file).	None	None	200 OK — route object. 404 Not Found — event or route does not exist.
POST	/api/events/{id}/route	Adds a route to an event.	Owner (Organiser)	{ routeName, distanceKm, elevationGainM, startLocation, gpxFilePath }	201 Created — route object. 403 Forbidden — not the event owner.
GET	/api/events/{id}/weather	Returns the live/forecast weather for the event's date and location, to help participants prepare for race day.	None	None	200 OK — weather summary object. 404 Not Found — event does not exist.

Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}/categories	Lists all categories (distances/divisions) for an event.	None	None	200 OK — array of category objects. 404 Not Found — event does not exist.
POST	/api/events/{id}/categories	Adds a new category to an event.	Owner (Organiser)	{ categoryName, routeId, distanceKm, entryFee, maxParticipants }	201 Created — category object. 403 Forbidden — not the event owner.
PUT	/api/categories/{id}	Updates a category's details.	Owner (Organiser)	{ categoryName, distanceKm, entryFee, maxParticipants }	200 OK — updated category object. 403 Forbidden. 404 Not Found.
DELETE	/api/categories/{id}	Removes a category from an event.	Owner (Organiser)	None	204 No Content. 403 Forbidden. 404 Not Found — category does not exist.

Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/categories/{id}/register	Enters the logged-in participant into a category, generating a bib number.	Participant	{ paymentReference } (optional)	201 Created — registration object with bib number. 400 Bad Request — category full. 409 Conflict — already registered.
GET	/api/events/{id}/registrations	Lists everyone registered for an event, for organiser race-day management.	Owner (Organiser)	None	200 OK — array of registration objects. 403 Forbidden — not the event owner.
GET	/api/registrations/{id}	Returns a single registration's details.	Owner (Organiser) or owning Participant	None	200 OK — registration object. 403 Forbidden. 404 Not Found.
DELETE	/api/registrations/{id}	Withdraws/cancels a registration.	Participant (own registration)	None	204 No Content. 403 Forbidden — not the registration owner. 404 Not Found.

Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/registrations/{id}/result	Captures a finish result against a participant's registration.	Owner (Organiser)	{ finishTime, overallPosition, categoryPosition, resultStatus }	201 Created — result object. 400 Bad Request — result already captured. 403 Forbidden — not the event owner.
GET	/api/events/{id}/results	Returns the full results list/leaderboard for an event, optionally filtered by ?categoryId=.	None	None	200 OK — array of result objects. 404 Not Found — event does not exist.
PUT	/api/results/{id}	Corrects an existing result (e.g. after a timing dispute).	Owner (Organiser)	{ finishTime, overallPosition, categoryPosition, resultStatus }	200 OK — updated result object. 403 Forbidden. 404 Not Found.

Notes on deviations

Route and Weather endpoints are not explicitly named as a required section in the brief, but are included because the Part 2 functional requirements state participants must be able to "prepare for race day using live weather and route information" — these endpoints exist to serve that requirement and are nested under /api/events/{id}/... since they belong conceptually to an event.

"Owner (Organiser)" role checks mean the API must verify the logged-in organiser's UserID matches the event's OrganiserID (or the event linked to the category/registration/result being modified), not just that the caller holds the Organiser role generally.